

What problem have you noticed?

Start with life, not technology.

WHY THIS MATTERS

Great capstone ideas often begin with something ordinary that is annoying, slow, expensive, confusing, unsafe, repetitive, or unfair. Do not begin with 'I want to use AI'. Begin with 'I keep seeing this problem'.

GO AND DO THIS

1. Look at your own week: home, renting, transport, university, work, sport, health, shopping, research.
2. Notice moments where someone says: 'This takes forever', 'Why is this so hard?', 'I keep forgetting', or 'There must be a better way'.
3. Write down three problems before choosing one.
4. Prefer a problem you can observe and ask people about.

EXAMPLE

Example: A student looking for a rental checks several websites, repeats the same information, misses inspection times, and cannot compare properties easily. The idea is not 'build a rental app' yet. The observed problem is fragmented, repetitive rental searching.

RESEARCH LENS

Research project? Your 'problem' may be a scientific limitation: a method is inaccurate, a dataset is hard to use, a workflow is manual, or evidence is scattered across papers.

A. Capture three problems you have actually seen

Problem 1

What happened? Where did you see it? Keep it concrete.

Problem 2

What happened? Where did you see it? Keep it concrete.

Problem 3

What happened? Where did you see it? Keep it concrete.

B. Choose ONE problem to explore

My chosen problem in one sentence

Use: "When _____ tries to _____, they struggle because _____."

☐ I have personally observed this problem.

☐ I know people I can talk to about it.

Who actually feels this pain?

A problem without a person is only an assumption.

THINK LIKE AN ENTREPRENEUR

Your job is to identify a real person or group, not a vague market such as 'everyone'. Describe the person, the situation they are in, and what they are trying to achieve.

GO AND TALK TO PEOPLE

Talk to at least **three people** who experience the problem. Do not sell your solution. Ask about the last time the problem happened. Useful prompts: What were you trying to do? What happened? What was hardest? What did you do instead? How often does this happen? What does it cost you?

EXAMPLE

Example user: International student, 22-30, new to Brisbane, searching for a room while studying. They may have limited local knowledge, a tight budget, and only a few days to inspect properties.

RESEARCH LENS

Research project? The 'person' can be a researcher, clinician, analyst, laboratory, government team, or community that depends on the method or evidence.

Primary person / user

Be specific: role, situation, location/context, and what they are trying to achieve.

What did real people tell you?

Interview	Who / context	What they tried to do	Pain in their own words
Person 1			
Person 2			
Person 3			

What pattern did you hear across the conversations?

Do not write what you hoped to hear. Write what repeated.

How painful and important is the problem?

Measure the pain before you build the cure.

WHY THIS MATTERS

Some problems are real but not important enough to justify a capstone product. Estimate how often the problem happens and what damage it causes: time, money, mistakes, stress, missed opportunities, safety, or quality.

FIND EVIDENCE

Search for supporting evidence. Use credible statistics, reports, papers, forums/reviews, organisational data, or your own interviews. Triangulate: one source is a clue; several independent sources are evidence.

TURN FEELING INTO EVIDENCE

Example: 'Rental search is annoying' is weak. Stronger: 'Three interviewed students each spent 6-10 hours per week searching; two missed inspections because information was spread across platforms.'

RESEARCH LENS

For research, quantify the scientific pain: runtime, error rate, sample size limitation, reproducibility issue, missing benchmark, or evidence gap.

How often does it happen?

Per day/week/month? To whom? In what situations?

What does it cost?

Estimate time, money, errors, frustration, delay, risk, or missed opportunity.

Evidence you found

#	Source / link / person	What evidence does it provide?
1		
2		
3		

Why is this problem worth solving?

Finish: "This matters because..."

What do people do today?

Your real competitor may be Excel, WhatsApp, or doing nothing.

LOOK BEYOND DIRECT COMPETITORS

Before inventing a solution, understand the current behaviour. Search products, apps, services, papers, open-source tools, spreadsheets, manual processes, and workarounds.

DO THE SEARCH

Find at least three alternatives. For each one, actually inspect it: website, demo, reviews, documentation, paper, GitHub repository, or user feedback. Ask: what job does it do well, and where does the user still struggle?

EXAMPLE

Example: For rental discovery, alternatives might include Domain, realestate.com.au, Facebook groups, property agents, and a personal spreadsheet. A manual workaround can be more important than a software competitor.

RESEARCH LENS

For research projects, compare at least three papers/methods/tools/datasets that currently address the same scientific task.

Investigate three existing systems / approaches

Alternative	What it does well	Where users struggle	Evidence / link
System 1			
System 2			
System 3			

What is the current manual workaround?

What do people do when none of the systems fully solve the problem?

Why are existing solutions not enough?

Find the gap. Do not invent one.

THE GAP IS YOUR OPPORTUNITY

Now compare what you learned. A useful gap is something users care about that existing solutions do poorly, inconsistently, expensively, or not at all.

HOW TO USE SWOT

Use SWOT as a thinking tool, not a decoration.

Strengths: why people use current solutions.

Weaknesses: where they fail users.

Opportunities: unmet needs or new capabilities.

Threats: reasons your idea may fail - regulation, strong incumbents, data access, switching cost.

EXAMPLE

Example gap: Existing rental platforms have many listings (strength) but do not combine a student's classes, commute time, inspection schedule, and application readiness (weakness/opportunity). A threat is that platforms already own the listing data.

RESEARCH LENS

In research, a gap should be supported by literature: missing comparison, poor generalisation, inaccessible workflow, insufficient dataset, or an unanswered question.

SWOT each of the three systems / approaches you investigated

System	Strengths	Weaknesses	Opportunities	Threats

The gap I can defend with evidence is...

Use: "Existing approaches do _____ well, but users still struggle with _____ because _____."

What opportunity will you pursue?

Make one clear promise to one clear user.

FOCUS THE OPPORTUNITY

A strong concept has a narrow first promise. It does not try to fix an entire industry. Define the outcome you will improve and why your approach is meaningfully different.

FIVE QUESTIONS

Ask yourself:

1. Which user pain will I address first?
2. What outcome will improve?
3. Why is this better than the current workaround?
4. What will I deliberately NOT solve?
5. Why is now a reasonable time to attempt it?

EXAMPLE

Example value proposition: 'For international students searching for rentals, we will create a planning assistant that combines listings, commute, inspections and application tasks so they can make decisions with less repeated work.'

RESEARCH LENS

For research: state the research question or hypothesis, the knowledge gap, and the expected contribution - method, benchmark, dataset, evidence, or tool.

Who is it for?

Name the primary user, not everyone who could possibly use it.

What outcome will improve?

Examples: reduce time, reduce errors, improve access, improve prediction, simplify a workflow.

My one-sentence value proposition

For [user], who [pain], our [solution category] helps [outcome], unlike [current alternative], by [difference].

What will I NOT solve in this capstone?

A clear boundary makes the project stronger.

Why now?

New data? New API? New user behaviour? New research opportunity? A newly visible pain?

What exactly will the user experience?

Describe the journey before listing features.

TELL THE USER STORY

Imagine one real user using your solution from beginning to end. What triggers them to use it? What do they give the system? What happens? What useful output do they receive? What do they do next?

A SIMPLE TEST

Avoid technical words first. If you cannot explain the experience without saying 'LLM', 'database', or 'API', the product idea may still be unclear. Technology should serve the journey.

EXAMPLE

Example: Student opens the tool -> imports/saves candidate rentals -> enters class/work locations -> tool ranks options and builds an inspection plan -> student compares trade-offs and tracks applications.

RESEARCH LENS

For a research capstone, the journey may be: input dataset -> preprocessing -> method/analysis -> validation -> interpretable output -> researcher decision.

The trigger

What happens immediately before the user needs your solution?

INPUT - what does the user/system provide?

Information, files, sensor data, text, records, parameters, datasets...

PROCESS - what useful work happens?

Describe in plain English first.

OUTPUT - what does the user receive?

Recommendation, prediction, plan, dashboard, report, alert, result...

ACTION - what can the user do better afterwards?

This is the value your project creates.

What is the smallest useful version you can build?

An MVP proves value; it does not prove you can code everything.

THINK MVP

The capstone is not a competition to build the most features. Build the smallest version that lets you test whether the core idea helps the user.

THE TWO-WEEK TEST

Imagine you had only two weeks. What must exist for a user to experience the core value? Put everything else in 'later'. A feature belongs in the MVP only if removing it would make the core test impossible.

EXAMPLE

Example: MVP: save 10 rental options, calculate commute time, compare price/distance, create an inspection checklist. Later: automatic applications, landlord messaging, prediction models, payments.

RESEARCH LENS

For research: MVP may be a baseline pipeline + one proposed improvement + a reproducible evaluation on a manageable dataset, not the final publication-scale system.

Must-have MVP capabilities

MVP capability 1

What user value does this capability enable?

MVP capability 2

What user value does this capability enable?

MVP capability 3

What user value does this capability enable?

MVP capability 4

What user value does this capability enable?

Later - valuable, but not required for the first test

Future features

Park exciting ideas here so they do not expand your current scope.

MVP success sentence

"If a user can _____ and it improves _____, the MVP has done its job."

What information or data does your solution need?

No data plan = no believable system plan.

DATA IS A DEPENDENCY

List every important input your solution depends on. Ask where it comes from, whether you are allowed to use it, how reliable it is, how much you need, and what happens if it is unavailable.

CHECK BEFORE YOU PROMISE

For each source check: owner, access method, format, update frequency, cost, licence/terms, privacy, consent, missing data, bias, and whether a sample is available now.

EXAMPLE

Example: Rental listings may be visible on a website but scraping or reusing them may be restricted. A better MVP might use user-entered links, an approved API, or a small permitted dataset.

RESEARCH LENS

For research, record dataset name, cohort/sample size, labels/ground truth, access approval, train/test split, benchmark compatibility, and reproducibility constraints.

#	Data / information needed	Where will it come from?	Access / privacy / quality issue
1			
2			
3			
4			

What data can you access TODAY for a prototype?

Be realistic. Name the actual sample, public dataset, mock data, interview data, or permitted source.

What is the biggest data risk?

What could make the idea impossible or force a redesign?

How will you build it?

Choose technology only after the need is clear.

TECHNOLOGY SERVES THE IDEA

Now translate the user journey into components. Choose the simplest technology that can deliver the MVP. Every technology should have a reason to exist.

JUSTIFY EACH COMPONENT

Ask: Do I need a web/mobile interface? A database? Authentication? External APIs? Maps? Payments? Notifications? AI/ML? A research compute pipeline? Cloud deployment?

If you propose AI, explain what task requires it and what a simpler rule-based approach cannot do well enough.

EXAMPLE

Example stack: React/Next.js interface; PostgreSQL for saved rentals/tasks; Google Maps API for commute time; simple ranking formula first. AI might later summarise listing text - it is not required for the core ranking MVP.

RESEARCH LENS

Research path: Python/R, notebooks/pipeline, packages, compute/HPC/GPU needs, model/library dependencies, reproducibility and version control.

Component	Technology / service	Why is it needed?
Interface		
Backend / logic		
Database / storage		
External APIs		
AI / ML (if needed)		
Hosting / compute		

Draw the architecture in words

Example: User -> web app -> backend -> database/API -> result. Include where data enters and leaves.

☐ AI is not required for the MVP.

☐ AI is required and I can justify the task.

Can you realistically deliver and test it?

A good idea fits the time, skills, access and evidence you actually have.

BE AMBITIOUS, NOT IMPOSSIBLE

Feasibility is part of innovation. Identify what could break the project before it breaks the project. The goal is not to eliminate risk; it is to expose and manage it.

LOOK FOR PROJECT-KILLERS

Consider: technical complexity, unfamiliar skills, API/data access, privacy/security, ethics, cost, performance, external approvals, integration difficulty, user recruitment, and time.

EXAMPLE

Example risk: 'We depend on an API that has not approved access.'
Mitigation: prototype with a permitted sample or user-provided data.
Kill-test: confirm API terms in Week 1.

RESEARCH LENS

Research risks may include compute limits, unavailable labels, small sample size, data leakage, benchmark mismatch, irreproducible baselines, or ethics/data governance approvals.

Top risks and what you will do about them

#	Risk	H/M/L	Mitigation or early kill-test
1			
2			
3			
4			

How will you know the solution works?

Primary success metric

Pick something observable: time saved, task success, error reduction, accuracy, user rating, completion rate, benchmark improvement...

Who/what will you test it on?

Users, interviews, benchmark dataset, simulation, usability test, baseline comparison...

- ☐ Feasible in the capstone timeframe.
- ☐ Scope must be reduced before building.

Should this idea move forward?

Earn the right to build.

ENTREPRENEURS CHANGE THEIR MINDS

You have now observed, interviewed, measured, searched, compared, scoped, and planned. Your decision should be based on what you learned - not on how attached you are to the first idea.

MAKE AN EVIDENCE-BASED DECISION

A strong outcome can be:

Proceed - evidence supports the idea.

Refine - same problem, narrower/better solution.

Pivot - interviews revealed a different, stronger problem.

Stop - problem is weak, solved already, or infeasible. Stopping early is good judgement.

PITCH IT SIMPLY

60-second pitch structure: 'We spoke to [people] who struggle with [problem]. Today they use [alternatives], but [gap]. We propose [MVP] that improves [outcome]. We can build it using [key technology/data] and test success by [metric].'

RESEARCH LENS

For research: include the question, literature gap, proposed method/analysis, dataset, baseline, evaluation metric, and expected contribution.

My 60-second evidence-based pitch

Use the structure on the left. A reader should understand the problem, gap, solution and proof plan without another page.

The strongest evidence supporting this idea

What fact, interview pattern, comparison, or literature finding gives you confidence?

The biggest assumption still unproven

What must you learn next?

Decision

- ☐ PROCEED - build the MVP.
- ☐ REFINED - narrow/change the solution.
- ☐ PIVOT - pursue a better problem/gap.
- ☐ STOP - evidence does not justify building.

Next action I will take

One concrete action, owner, and target date.